Whole-Brain fMRI Decoding of Subjective Food Value: Cross-Participant Generalization and Spatial Attribution

Abstract

Subjective food value links food perception to eating choices, and related information spans visual, gustatory, and valuation systems. Decoding asks which of these systems carry value information, and applying a decoder to individuals excluded from its training reveals whether that information is shared. However, most value decoding, particularly for food, has been performed within participants or in predefined valuation regions. It therefore remains unclear which other regions carry food-value information that generalizes across individuals. Using a public food-valuation fMRI dataset (N = 31), we decoded high versus low value from the whole-brain activity of participants excluded from training. A linear support vector machine (SVM) and a three-dimensional convolutional neural network (CNN) reached balanced accuracies of up to 63.8% and 57.2%, respectively. Computed without predefined regions of interest, the SVM forward pattern and the CNN Layer-CAM maps converged on early and ventral visual cortex and the right posterior parahippocampal gyrus, and also involved posterior-medial, motivation-related subcortical, and classic valuation regions. This agreement between linear and nonlinear decoders, each interpreted with a different attribution method, indicates that the information supporting cross-participant decoding of food value extends beyond the classic valuation network.

Introduction

Subjective food value is a key link between sensory input and eating choices. Frameworks of value-based decision making describe how individuals represent the available options, assign values to them, and select actions accordingly (Rangel et al., 2008), and food value can be predicted from an individual's subjective beliefs about nutritive attributes (Suzuki et al., 2017). When a person sees a food photograph, the cues it provides must therefore be related to experience and current needs, and the current desire to eat offers a behavioral index of this process. Understanding how food-value information is distributed in the brain, and to what extent it is shared across individuals, is important for explaining how sensory information, internal needs, and choice behavior jointly shape eating decisions, and it provides a basis for understanding altered value processing in disordered eating.

Neuroimaging has linked food perception and valuation to several brain systems. Ventral visual cortex contains regions with strong selectivity for food images (Jain et al., 2023; Khosla et al., 2022; Pennock et al., 2023). Viewing food can also elicit taste-quality information in gustatory insular cortex (Avery et al., 2021). A broader literature relates ventromedial prefrontal, orbitofrontal, and ventral striatal activity to subjective value (Bartra et al., 2013). In addition, associative memory can bias later preferences (Wimmer & Shohamy, 2012), and posterior-medial and medial-temporal systems provide a framework for understanding context representation and memory-guided behavior (Ranganath & Ritchey, 2012). These studies separately explain the visual category, associated sensory attributes, value, and memory processes of food, but they do not directly address which spatial components appear together when a model predicts food value in held-out individuals.

Multivariate analyses can read food-evaluation information from patterns of brain activity. For example, Suzuki et al. (2017) decoded food value from within-participant orbitofrontal patterns, and Pogoda et al. (2016) examined how food-preference patterns generalized from chocolate bars to salty snacks. Cross-participant studies further showed that prefrontal value patterns predict preferences across individuals (Gross et al., 2014) and that orbitofrontal valence patterns classify the affect evoked by scenes and tastes across individuals (Chikazoe et al., 2014); because these analyses were restricted to predefined regions, they could not show where in the brain the value information shared across people is distributed. Another line of work trains whole-brain signatures on group data to predict negative affect or craving intensity in new individuals (Chang et al., 2015; Koban et al., 2023); in the craving signature, the predictive inputs were each participant's β maps summarized by rating level. These studies established that cross-individual prediction is feasible and highlighted a difference in what is being predicted: a map summarizing one rating level and the activity map of a single food evaluation contain different amounts of observation and noise. Cross-participant decoding is further constrained by individual differences in functional organization: in ventral temporal cortex, between-subject classification of anatomically aligned responses falls well below classification after functional alignment (Haxby et al., 2011). Whether subjective food value can be decoded from whole-brain activity summarized over one to a few trials, without training data from the test participant or functional alignment, therefore remains to be tested.

Such multivariate approaches require many stimuli and trials per participant to obtain stable feature representations and to avoid overfitting to noise (Sugawara et al., 2026). In rapid event-related designs, the hemodynamic

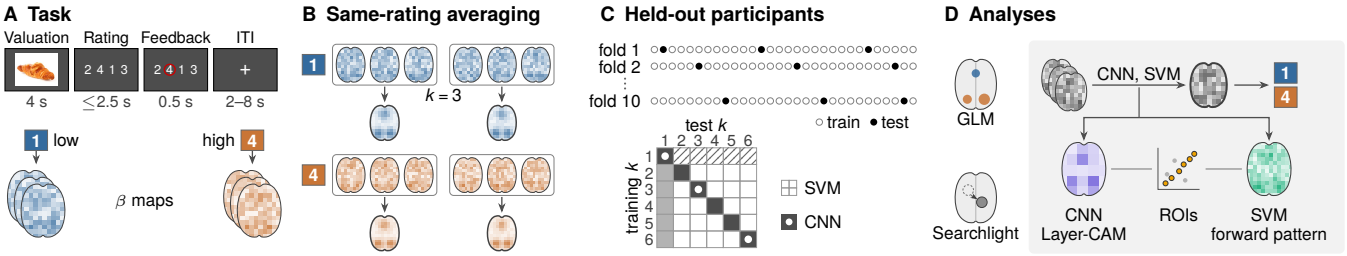


Figure 1: Study design. (A) Trial sequence. Ratings of 1 and 4 on the four-point desire-to-eat scale defined low and high value, and each valuation event yielded one single-trial β map (3,426 low and 3,612 high across the 30 decoding participants). (B) Within each participant and rating, β maps of different foods were averaged in nonoverlapping groups of k (k = 3 shown; five random groupings for each k = 2–6). (C) Ten-fold participant-held-out cross-validation (27 training and three test participants per fold). The SVM was trained and tested at every combination of k (shading: the conditions in Figure 2B), and the CNN at k = 1, 3, and 6 (dots). (D) Group-mean GLM (n = 31), within-participant searchlight, and cross-participant decoding; the CNN Layer-CAM and SVM forward patterns of the cross-participant decoders were compared across ROIs.

responses to adjacent trials overlap in time, which complicates estimating the signal unique to each trial (Mumford et al., 2012); preprocessing choices such as trial averaging can substantially affect decoding results while reducing the number of available samples (Grootswagers et al., 2017), and in simulations modest averaging improved decoding when the averaged pseudotrials served as both training and test data (Scrivener et al., 2026). Averaging could raise cross-participant accuracy because cleaner training inputs let the decoder learn a better readout, or because the same readout reads more reliable observations of the held-out person; which of the two limits single-trial readout has not been tested. We therefore varied the number of averaged β maps within the same rating and crossed the granularity of training and test inputs. Spatially, univariate brain mapping identifies regions whose mean activity changes across conditions, whereas multivoxel pattern analysis and searchlight ask where activity patterns carry information about the experimental condition (Haxby et al., 2001; Kriegeskorte et al., 2006); Layer-CAM relates a CNN's class predictions to specific spatial features (Jiang et al., 2021). Deep learning is advantageous only when nonlinear relationships in the data can be exploited at the available sample size; in age and sex prediction from brain images, simple linear models perform on par with more complex models (Schulz et al., 2020). We therefore also used a matched linear SVM.

We used a densely sampled, openly available food-valuation fMRI dataset in which participants rated each of 568 different food photographs once across three scanning days (Sugawara et al., 2026). After defining low and high value by the lowest and highest points of a four-point rating scale, we trained a CNN and a linear SVM for cross-participant prediction on whole-brain single-trial and averaged β maps. GLM and within-participant searchlight analyses characterized the local spatial evidence, Layer-CAM described the class attributions of the cross-participant CNN, input replacement tested whether the attributed voxels were necessary for prediction, and the forward pattern of the SVM was compared across k and with Layer-CAM. We thus tested whether subjective food value generalizes to individuals excluded from training at the granularity of single valuation events, whether aggregating observations of the same rating acts on the learned readout or on the observations it reads, and how the spatial attributions of the cross-participant model relate to existing interpretations of visual, gustatory, memory-related, and motivational brain regions.

Methods

Dataset, task, and value labels

We analyzed the public food-valuation dataset of Sugawara et al. (2026) (OpenNeuro ds007267). Thirty-one healthy native Japanese speakers (11 women; age 21.29 $\pm$ 1.87 years) refrained from food and non-water drinks for 3 h before scanning and evaluated the same 568 food photographs across three scanning days, each photograph once. Ratings expressed current desire to eat on a four-point scale with a randomized button-to-rating mapping (Figure 1A). The original study received ethics approval from Hitotsubashi University (2025C069), with written consent for participation and data sharing.

We used ratings 1 and 4 as low- and high-value labels. Throughout, subjective food value refers to value operationalized by these ratings of current desire to eat, as distinct from hedonic responses during tasting, long-term preferences, or actual intake (Berridge, 2009). Participant sub-016 completed only two runs on the third day

and was excluded from all multivariate analyses (n = 30); the univariate GLM retained all 31.

Preprocessing and single-trial estimates

Images were preprocessed with fMRIPrep 25.2.3 and registered to MNI152NLin2009cAsym space (Esteban et al., 2019). Run-wise least-squares-all models in SPM25 estimated a separate β map for each valuation event. Models included motion, aCompCor, DVARS, and outlier regressors, with a 128-s high-pass filter. These single-trial inputs were not smoothed further. Each input was standardized within its participant's brain mask using its own mean and SD, with zeros outside the mask.

Averaging within the same rating

We examined individual trials (k = 1) and averages of k = 2–6 trials (Figure 1B). Within each participant and observed rating, trials were shuffled and partitioned into nonoverlapping groups; incomplete final groups were discarded. Averaging did not require trials from the same image and never crossed participants or ratings. The decoder comparison (Table 1) used the groupings of random seed 42 at k = 3 (1,133 low, 1,193 high inputs) and k = 6 (559 low, 588 high); the granularity analyses used five groupings (seeds 42–46) at each k = 2–6. The averaged maps were then standardized as above. For $k > 1$, the predicted object is therefore the aggregated pattern of multiple observations of the same rating.

Cross-participant validation and models

Each k used ten-fold participant-held-out cross-validation (27 training and three test participants per fold; Figure 1C) without response-based functional alignment. Within each k, the CNN and SVM used identical inputs, labels, and folds. The granularity analyses used the k = 1 partition at every k and refitted only the SVM, reproducing the original predictions at k = 1, 3, and 6 exactly. Each fold's SVM trained at k_{train} was also tested at every k_{test} (same grouping seed).

The CNN comprised three 3D convolutional blocks with batch normalization and ReLU activations, followed by a 128-unit fully connected layer and a single logit output. Training minimized binary cross-entropy with Adam (learning rate and weight decay 10^{-4}), used class-balanced batches of 16, and ran for five fixed epochs with seed 42; the final checkpoint generated predictions for the held-out participants. The matched SVM was a linear support vector classifier (C = 1, balanced class weights) whose features were the standardized values within a fixed 222,441-voxel anatomical intersection mask. Neither model used hyperparameter search.

Performance and statistics

The primary performance measure was balanced accuracy (BA), the mean of the two class-specific recalls. We calculated BA from all test predictions of each held-out participant, averaged it with equal weight across the 30 participants, and reported the sample SD; 95% confidence intervals for mean BA were percentile intervals from 100,000 participant resamples. Performance against 50% used one-sided one-sample t tests, and differences between the two models at the same k used two-sided paired t tests, both unadjusted across k.

For the CNN, comparisons across k tested the participant BA differences $k = 3 - k = 1$, $k = 6 - k = 1$, and $k = 6 - k = 3$ on the seed-42 inputs. For the SVM granularity analyses, participant BA was averaged over the five groupings; each k was compared with k = 1 and with $k - 1$, and training and test granularity were compared at fixed test or training granularity. Each family used two-sided paired t tests with Holm correction and two-sided sign-flip tests (100,000 flips) as a check; a per-participant slope of BA on $\log_2 k$ summarized the trend. The between-participant SD expected from sampling alone was computed from each participant's own recalls r_1, r_4 and numbers of test inputs n_1, n_4 as $\sqrt{0.25\,[r_1(1-r_1)/n_1 + r_4(1-r_4)/n_4]}$.

Univariate and within-participant local pattern analyses

Condition-wise GLMs used BOLD images smoothed at 6-mm FWHM, with regressors for the four ratings and for task, motion, white-matter, cerebrospinal-fluid, and outlier terms. High-minus-low and low-minus-high contrasts were combined across runs by fixed effects and entered a group analysis of 31 participants with 5,000 sign flips, threshold-free cluster enhancement, and whole-brain family-wise error (TFCE-FWE) correction at $p \leq .05$ (Smith & Nichols, 2009; Winkler et al., 2014).

Within-participant whole-brain SVM and searchlight analyses used unsmoothed run-level condition β maps for ratings 1 and 4 and three-fold leave-one-session-out validation. Searchlights comprised 6-mm-radius spheres analyzed with a linear SVM (C = 1, balanced weights) at each center voxel (Kriegeskorte et al., 2006). Each participant's three-fold BA map, centered on chance, entered a one-sided group test of 30 participants with 5,000 sign flips, a cluster-forming threshold of $p < .001$, and cluster-extent FWE correction at $p \leq .05$. GLM clusters and searchlight coverage were labeled with the same atlas used for the Layer-CAM summaries.

Layer-CAM and regional summaries

Layer-CAM was calculated from the final k = 6 CNN for correctly classified held-out inputs (Jiang et al., 2021). The target was the output logit for high value and its negative for low value. Positive gradients of the target with respect to the final convolutional feature maps weighted those maps, followed by a ReLU and trilinear interpolation from the $7 \times 9 \times 7$ feature grid to the input volume.

Layer-CAM values denote attribution to the target class and do not indicate the direction of BOLD change. Maps were averaged first within each participant and class, then equally across contributing participants: low-value maps included 267 correct inputs from 27 participants and high-value maps 407 inputs from 29 participants; 26 participants contributed to both. For surface displays, participant maps entered a one-sample test against zero (5,000 sign flips, TFCE-FWE); because the nonnegative maps exceeded zero throughout the common mask, the displays show the highest 20% of t values within each class as a ranking of participant consistency.

Regional values were extracted from the unthresholded class-specific group mean volumes using the Harvard–Oxford maximum-probability cortical and subcortical atlases (25% threshold, 2-mm resolution). Each region's mean Layer-CAM value was divided by the whole-brain mean of the same group volume, giving a whole-brain reference of 1; normalization followed the group-first order. Five study-defined groups—occipital, parietal, temporal, frontal, and insular–subcortical—were summarized by weighting constituent ROI means by voxel count; cingulate cortex was assigned to the frontal group, and insular and subcallosal cortex to the insular–subcortical group. For participant-level inference, the same regional values were computed from each participant's class map normalized to its own whole-brain mean; two-sided sign-flip tests compared each class with 1 (27 and 29 participants), and paired sign-flip tests compared high with low value in the 26 participants contributing to both (100,000 flips; Holm correction within the five regional groups and within ten selected posterior-medial, medial-temporal, insular, and subcortical ROIs).

Input replacement and forward patterns

Input replacement tested whether attributed voxels were needed by the fixed $k = 6$ CNN: selected voxels were set to the whole-brain mean of the training-fold inputs, with model parameters unchanged. The conditions were the top 20% of training-fold Layer-CAM voxels, 100 equal-volume random masks, a value-related ROI set (frontal medial and orbital cortex and bilateral nucleus accumbens), a visual ROI set, and their union; four primary comparisons used one-sided sign-flip tests (100,000 flips) with Holm correction. Haufe-transformed forward patterns of the SVM (Haufe et al., 2014) were computed for every k and grouping and compared across k by Pearson correlation; the cosine similarity of fold-matched weight vectors compared the learned readouts. At matched granularity ($k = 6$), the mean magnitude of the forward pattern in 112 Harvard–Oxford ROIs (at least 30 voxels) was compared with the mean of the two Layer-CAM class maps by Spearman correlation, and the signed pattern described each class's direction of activity.

Table 1: Cross-participant decoding with matched inputs (seed-42 groupings). BA is participant-mean balanced accuracy ± sample SD across the same 30 held-out participants; Figure 2 gives the SVM at every k over five groupings.

k	Low/high	Inputs	CNN BA (%)	SVM BA (%)
1	3,426/3,612	7,038	53.71 ± 6.56	57.87 ± 5.96
3	1,133/1,193	2,326	54.58 ± 8.43	63.80 ± 10.33
6	559/588	1,147	57.24 ± 10.60	63.33 ± 15.76

Results

Subjective food value was decodable in held-out individuals

Both models predicted the high and low value of held-out participants in all three input settings (Table 1). With single-trial inputs, participant-mean BA was 53.71% ± 6.56% for the CNN and 57.87% ± 5.96% for the linear SVM (one-sided $p = .00213$ and $p = 2.83 \times 10^{-8}$); BA exceeded 50% in 27 of 30 participants for the SVM and in 25 of 30 for the CNN. The SVM had higher participant-mean BA than the CNN at every k, with paired differences of 4.16, 9.23, and 6.08 percentage points (all two-sided $p < .002$). The $k = 6$ CNN used for spatial attribution reached 57.24% (95% CI [53.38, 60.84]%; $t(29) = 3.745$, one-sided $p = .000398$).

Gross et al. (2014) proposed that prefrontal value signals follow a common scale that is comparable across individuals; here, value was decoded from single-trial whole-brain maps of individuals excluded from training.

Averaging observations of the same rating improved linear readout

With five random groupings per k on a common participant partition (Figure 2A), mean SVM BA was 57.87%, 61.06%, 63.14%, 63.01%, 63.47%, and 63.15% for $k = 1$–6. Every $k > 1$ exceeded single trials, by 3.19 to 5.60 percentage points (Holm $p \leq .016$), and BA rose by 2.13 points per doubling of k (95% CI [0.69, 3.45], $p = .006$; positive in 25 of 30 participants). The increase was concentrated in the first steps: $k = 2$ exceeded $k = 1$ by 3.19 points (Holm $p = .013$) and $k = 3$ exceeded $k = 2$ by 2.08 points (Holm $p = .005$; 26 of 30 participants), whereas differences among $k = 3$–6 were not detectable ($|\Delta| \leq 0.46$ points, Holm $p = 1$) at a point where each participant contributed fewer than 80 test inputs (38 at $k = 6$). Group means varied by 1.3 to 3.7 points across groupings at the same k, and the seed-42 sets of Table 1 lay within this range. On matched inputs, mean CNN BA also increased with averaging (53.71%, 54.58%, 57.24% at $k = 1, 3, 6$), but the participant-level differences were not significant (Holm $p \geq .19$).

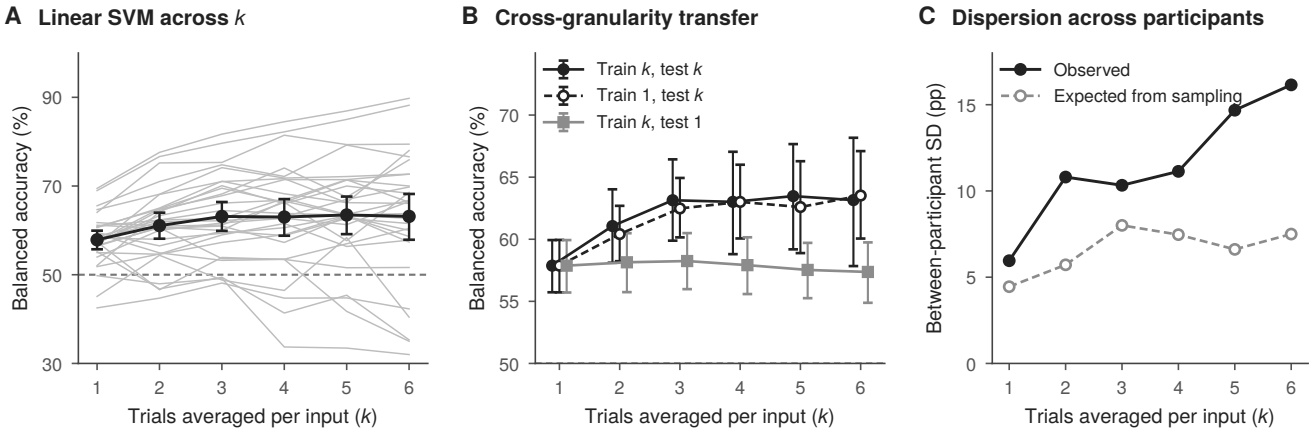


Figure 2: Observation granularity and cross-participant readout (linear SVM; $k = 1$ participant partition at every k; $k > 1$ averaged over five random groupings). (A) Balanced accuracy for $k = 1$–6; grey lines, participants; black, participant mean with bootstrap 95% CI; dashed line, chance. (B) Cross-granularity transfer: SVMs trained and tested at k, trained on single trials and tested at k, and trained at k and tested on single trials. (C) Between-participant SD of BA (one grouping, seed 42) and the SD expected from binomial sampling of each participant's own test inputs.

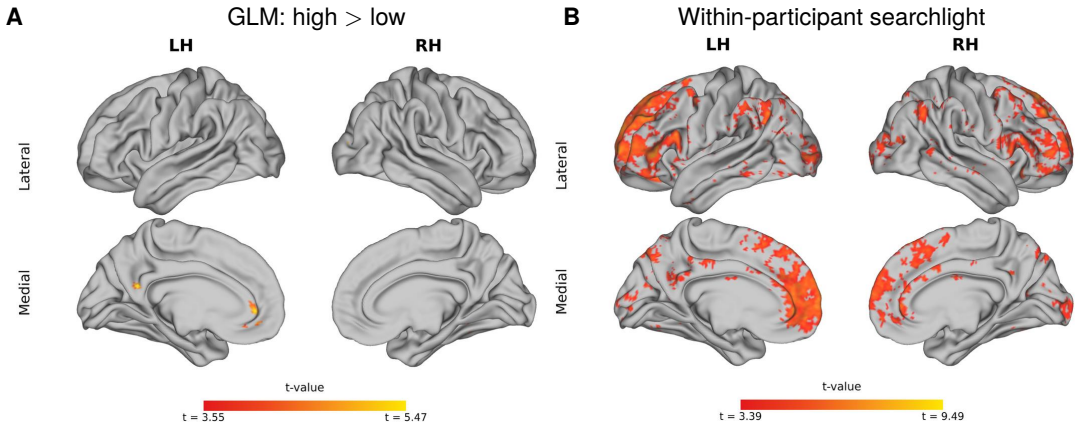


Figure 3: Univariate mean differences and within-participant local pattern decoding. (A) High-greater-than-low GLM result (31 participants; TFCE-FWE $p \leq .05$). (B) Within-participant 6-mm searchlight (30 participants; cluster-forming threshold $p < .001$, cluster-extent FWE $p \leq .05$). Colors represent group-level t values of each analysis on separate scales.

Averaging acted on the held-out observations, not on the readout

Averaging changes both the training and the test inputs, so we crossed the two (Figure 2B). An SVM trained on single trials and applied to averaged held-out inputs reached 60.42%, 62.48%, 63.00%, 62.59%, and 63.52% for $k = 2$–6, 2.54 to 5.65 points above single-trial testing (Holm $p < .001$; 24 to 27 of 30 participants). With training fixed in this way, BA rose from $k = 1$ to 2 and from 2 to 3 (Holm $p \leq .001$) and not detectably beyond (+1.04 points from $k = 3$ to 6, $p = .36$), so the absence of further gain did not come from fewer training inputs. At the same test granularity, training on averaged instead of single-trial inputs changed BA by -0.37 to +0.88 points (Holm $p = 1$), and SVMs trained on averaged inputs read single trials as well as the single-trial model (-0.51 to +0.37 points, Holm $p = 1$). The models learned at different k differed no more than models fitted to different groupings: their forward patterns correlated at $r \geq .99$ across k, and the cosine similarity of fold-matched weight vectors across k (.75 to .87) was similar to that between two random groupings at the same k (.84 to .85).

Averaging enlarged stable differences between participants

The between-participant SD of SVM BA grew from 5.96 points at $k = 1$ to 16.15 at $k = 6$ (one grouping), whereas binomial sampling of each participant's own test inputs predicted 4.45 to 8.00 points, so the dispersion in excess of sampling grew from about 4 to 14 points (Figure 2C). Participants kept their order across k (seed-averaged BA; $r = .71$ between $k = 1$ and 6, $r = .92$ between $k = 3$ and 6): at $k = 6$, nine participants reached at least 70% and five remained below 50%.

Univariate and within-participant local pattern analyses

The high-greater-than-low GLM contrast yielded 473 significant voxels after whole-brain TFCE-FWE correction, mainly in a ventromedial prefrontal region formed by frontal medial, paracingulate, and anterior cingulate cortex, with additional clusters in posterior cingulate, right temporal occipital fusiform cortex, occipital cortex, and cerebellum; the reverse contrast yielded none (Figure 3A). The dataset paper replicated subjective-value encoding in ventromedial prefrontal cortex and ventral striatum (Sugawara et al., 2026); our result agrees in ventromedial prefrontal cortex, whereas ventral striatum did not reach significance. Within-participant searchlight decoding identified significant clusters covering 31,431 voxels (highest local BA 67.36%; Figure 3B), with the greatest coverage in frontal cortex (36.6% of frontal ROI center voxels), followed by occipital (20.3%) and parietal (15.1%) cortex, and almost none in subcortical structures; within-participant whole-brain SVM decoding reached 62.41% $\pm$ 13.26% BA (two-sided $p = 1.78 \times 10^{-5}$). Paracingulate, anterior cingulate, and middle frontal gyri were extensively covered, consistent with Pogoda et al. (2016), who distinguished preferred from non-preferred foods in medial prefrontal, dorsal anterior cingulate, and dorsolateral prefrontal cortex.

Regions highlighted by Layer-CAM: prior interpretations and gaps

The low- and high-value Layer-CAM maps both extended across occipital, posterior-medial, temporal, and parts of frontal cortex and were spatially similar (Figure 4A–B; Pearson $r = .711$). Occipital values were the highest in both classes (low/high 1.254/1.350), followed by the insular–subcortical group (1.210/1.238), whereas parietal, temporal, and frontal groups were below the whole-brain reference of 1 (0.943/0.879, 0.819/0.895, and 0.904/0.784). Relative to low value, the high-value map placed more attribution on the occipital and temporal groups and less on the frontal and parietal groups, and the insular–subcortical group did not differ between classes (participant-level paired tests, Holm-corrected).

Table 2 relates each region highlighted by Layer-CAM to its prior interpretation and to the question that interpretation leaves open.

Input replacement and the SVM forward pattern

Replacing any voxel set changed the BA of the fixed CNN only slightly, from 57.24% to between 55.42% and 55.93%, and CAM-selected voxels were no more critical than equal-volume random voxels (Holm $p = .552$), as in the "virtual lesion" analysis of the negative-affect signature, in which no subsystem was necessary or sufficient for prediction (Chang et al., 2015). Only the value-related ROI set gave a reduction that survived correction (1.57 points, 95% CI [0.46, 2.78], Holm $p = .023$); it covered 4,092 voxels, whereas the random masks covered 44,496 voxels and reduced BA by a similar 1.61 points. Because classifier weights do not by themselves indicate signal origin, we compared the forward pattern of the SVM with Layer-CAM at matched granularity ($k = 6$). Across 112 ROIs, the magnitude of the forward pattern and the mean Layer-CAM value were ordered similarly (Spearman $\rho = .65$), and 13 ROIs ranked among the top 20 of both, against 3.6 expected by chance (Figure 4C–D): bilateral intracalcarine, supracalcarine, cuneal, and lingual cortex; the right occipital pole, occipital fusiform gyrus, inferior lateral occipital cortex, and temporal occipital fusiform cortex; and the right posterior parahippocampal gyrus, the only one outside the occipital group. The right posterior parahippocampal gyrus also carried a high-value emphasis in both models (forward pattern $z = +0.90$; Layer-CAM high $>$ low, Table 2). Bilateral precuneus, posterior cingulate, accumbens, thalamus, and caudate ranked above the median in both, whereas the right pallidum, putamen, and insula ranked high only for Layer-CAM. In the signed pattern, activity in the right posterior parahippocampal gyrus, posterior cingulate, left precuneus, accumbens, caudate, left anterior cingulate, and left thalamus was relatively higher with high value, and activity in early visual cortex relatively higher with low value. As the forward patterns were highly stable across k ($r \geq .99$), this profile did not depend on observation granularity.

Discussion

Cross-participant readout and observation granularity

Crossing the granularity of training and test inputs showed how averaging acted. A readout learned from single trials already carried the information that generalized across individuals, models trained on averaged inputs learned an almost identical readout, and averaging raised accuracy mainly by making the held-out observations more reliable. For signatures trained on rating-level condition maps, this suggests that the reliability of the observations to which they are applied matters

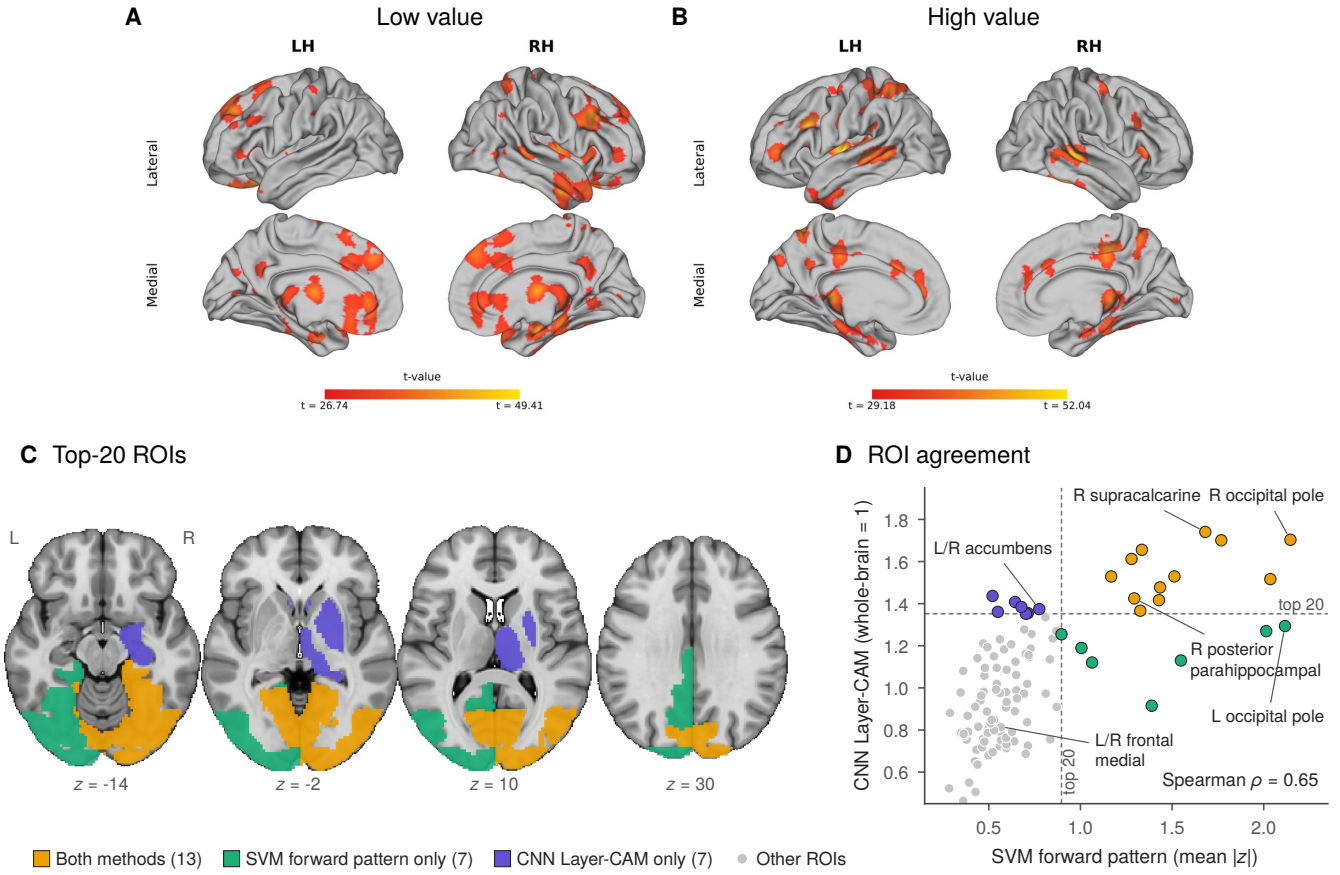


Figure 4: Spatial attribution of the cross-participant decoders. CNN Layer-CAM attributions for correct out-of-fold predictions of (A) low and (B) high value: highest 20% of participant-consistency *t* values, each class on its own scale. (C, D) Agreement between Layer-CAM (mean of the two class maps; whole-brain mean = 1) and the SVM forward pattern (k = 6, mean $|z|$; whole-brain mean 0.63) across 112 Harvard–Oxford ROIs. (C) ROIs ranked among the top 20 by both methods, by the forward pattern only, or by Layer-CAM only. (D) Forward-pattern magnitude against Layer-CAM value per ROI; dashed lines mark the 20th-ranked value on each axis, so the colored quadrants correspond to C. Thirteen ROIs ranked in the top 20 of both methods (3.6 expected by chance).

more than their training granularity. The averaging that helped was modest: k = 6 corresponds to about 5% of the median participant's trials per class, within the range that improved decoding in simulations (Scrivener et al., 2026). As observation noise fell, differences between participants grew beyond sampling and kept their order, consistent with the individual differences in functional organization that limit cross-participant decoding of anatomically aligned responses.

What drives prediction across individuals?

Because the decoder was evaluated on participants excluded from training, it reads the component of subjective value that is shared across individuals. The attributions of the cross-participant model were most prominent in visual cortex. Layer-CAM came from the k = 6 CNN, whose inputs averaged the β maps of six different foods with the same rating; that occipital cortex still carried the most attribution suggests that foods a person rates alike share visual properties, such as the color and form through which ventral food streams represent food, or food-category information beyond such features (Table 2). The forward pattern of the SVM showed the same emphasis and the same right posterior parahippocampal peak, so the profile is not specific to the CNN or to Layer-CAM. Classic valuation regions ranked low in this profile, yet replacing the value-related ROI set reduced performance as much as replacing a random set more than ten times larger, and the high-value direction of the forward pattern included the anterior and posterior cingulate, paracingulate, and frontal medial cortex of the GLM clusters. The GLM, within-participant searchlight, and cross-participant models address three levels—group mean differences, within-participant local readability, and the readout in

Table 2: Regions highlighted by Layer-CAM, prior interpretations, and open questions. Values: mean Layer-CAM (whole-brain mean = 1), low/high value; parentheses: class differences that passed participant-level paired tests (26 participants, Holm-corrected).

Region	Layer-CAM result	Prior interpretation	Open question
Visual cortex	Occipital group 1.254/1.350 (high $>$ low); top ROI for low value: R supracalcarine 1.980; for high value: R occipital pole 1.771	Food selectivity of ventral visual cortex is not explained by low- to mid-level features (Khosla et al., 2022); color and form jointly represent food (Pennock et al., 2023); the food region overlaps less across individuals after anatomical alignment (Jain et al., 2023)	These studies compared food with other image categories; whether visual responses reflect how much a person wants a given food was not tested
Insula	R insula 1.242/1.218 (no class difference)	Viewing food pictures retrieves the taste quality of the depicted food in dorsal mid-insula; picture- and taste-evoked patterns were unrelated (Avery et al., 2021)	Taste categories were decoded; how taste-related information relates to the desire to eat is untested
Posterior-medial and medial-temporal	R posterior PHG 1.296/1.554 (high $>$ low); R hippocampus 1.352/1.464; R precuneus 1.546/1.125 (low $>$ high); R posterior cingulate 1.252/1.218	Posterior-medial–parahippocampal system supports context and memory-guided cognition (Ranganath & Ritchey, 2012); hippocampus spreads reward value across associated memories (Wimmer & Shohamy, 2012)	Evidence comes from a memory review and monetary-reward learning; the role of experience and context in food valuation is unclear
Subcortical, motivation-related	Accumbens L 1.436/1.274, R 1.482/1.266 (R: low $>$ high); R thalamus 1.265/1.504 (high $>$ low); pallidum, putamen, amygdala $>$ 1 in both	Wanting and liking are separable, with hedonic hotspots in accumbens and ventral pallidum (Berridge, 2009); accumbens cue responses relate to intake but not to self-reported hunger, wanting, or liking (Lawrence et al., 2012); anterior insula and thalamus show mixed-sign value effects, possibly U-shaped (Bartra et al., 2013)	The roles of these regions in high- and low-value judgments defined by ratings are unclear
Classic valuation regions	Frontal group 0.904/0.784 (low $>$ high); value-related ROI replacement -1.57 points	Value signals in ventromedial prefrontal, orbitofrontal, and ventral striatal regions come mainly from within-participant analyses; the craving signature includes ventromedial prefrontal and cingulate cortex, ventral striatum, and mediodorsal thalamus, with rating-level condition maps as inputs (Koban et al., 2023)	How much single-trial cross-participant classification depends on these regions is unclear

held-out participants—so their spatial rankings need not agree.

The open questions listed in Table 2 can be tested directly: modeling the color and nutrient values provided for each image (Sugawara et al., 2026), together with shape and category, would separate early visual features from food-category information in occipital attribution, and manipulating the familiarity of foods or hunger and satiety would test whether medial-temporal, posterior-medial, and subcortical attributions follow a person's own rating. The analyses decoded only the rating endpoints, so whether the readout extends to intermediate ratings remains to be tested. The CNN used one architecture, one seed, and five fixed epochs, so the SVM advantage applies to this configuration rather than to nonlinear decoders in general. Inference relied on saved out-of-fold predictions rather than permutation tests with full retraining, and attributions were computed for correctly classified inputs only.

Conclusion

Subjective food value can be decoded from the whole-brain activity of individuals excluded from training. Interpreted without predefined regions, linear and nonlinear decoders converged on visual and parahippocampal regions in addition to classic valuation regions, and averaging observations of the same rating improved this readout mainly by making test observations more reliable. Testing the same foods before and after selective satiation would show whether this readout follows a person's value updating.

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